

PAPER_1606 — Bottom Quark Mass = $m_b = D + F \cdot D - F \cdot SSQ - F^2 \cdot D_{crit} + F^2 \cdot D_{BSFG} + F^2 \cdot D - F^2 \cdot SSQ^2 - F^2 \cdot SSQ^3 \approx 4.178 \text{ GeV}$

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Particle Physics **Date:** June 18, 2026 **Status:** CLOSED — 0.050% residual

Observation

PAPER_1209HH_S657 (Particle Physics Unified Proof Set) gives the closed-form expression:

$$m_b = D + F \cdot D - F \cdot SSQ - F^2 \cdot D_{crit} + F^2 \cdot D_{BSFG} + F^2 \cdot D - F^2 \cdot SSQ^2 - F^2 \cdot SSQ^3 \approx 4.178 \text{ GeV}$$

Residual to CODATA: **0.050%**.

UQFF Closed Identity

$$m_b = D + F \cdot D - F \cdot SSQ - F^2 \cdot D_{crit} + F^2 \cdot D_{BSFG} + F^2 \cdot D - F^2 \cdot SSQ^2 - F^2 \cdot SSQ^3 \approx 4.178 \text{ GeV} \quad 0.050\%$$

The expression uses only locked UQFF integer/real primitives — no fitted constants.

NOT REPLACEMENT

Experimental particle physics measures this constant directly. UQFF supplies a structural derivation from the integer primitives.

Reference

- Source: PAPER_1209HH_S657
- Related: PAPER_1494-1605 (other session-2026-06-18 closures)
- Calculator dispatch: `calculate_paradox({"paradox": "m_b_bottom_4_18"})`

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